PROBABILITY AND STATISTICS I

Assignment 1

Counting techniques — multiplication rule, permutations, combinations

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Name: ..... ID: .....

Group: ..... Date: .....

> **Before you start.** For every question, write one line naming the model before you write a formula: **does order matter**, and **may an item repeat**. An unjustified correct number earns less than a justified one. Show the cancellation — do not hand in a calculator display.

Q1 — Select the correct answer

1- Sarah goes to her local pizza parlour and orders a pizza. She can choose either a large or a medium pizza, has a choice of seven different toppings, and can have three different choices of crust. How many different pizzas could Sarah order?

A) 12 **B)** 20
C) 27 **D)** 42

2- Derek must choose a four-digit PIN number. Each digit can be chosen from 0 to 9. How many different possible PIN numbers can Derek choose?

A) 5040 **B)** 6561
C) 9000 **D)** 10000

3- For her literature course, Rachel has to choose one novel to study from a list of four, one poem from a list of six and one short story from a list of five. How many different choices does Rachel have?

A) 15 **B)** 120
C) 225 **D)** 240

4- James has to choose his options in his final year at school. He must choose one subject from each of five different option groups. In the first group there are six subjects to choose from; in the second, four; in the third, five; in the fourth, two; and in the fifth, four. How many possible choices of subjects can James make?

A) 480 **B)** 960
C) 1200 **D)** 4800

5- Jenny has nine different skirts, seven different tops, ten different pairs of shoes, two different necklaces and five different bracelets. In how many ways can Jenny dress up?

A) 6300 **B)** 7560
C) 12600 **D)** 63000

6- How many permutations of 3 different digits are there, chosen from the ten digits 0 to 9 inclusive?

A) 84 **B)** 120
C) 504 **D)** 720

7- How many permutations of 4 different letters are there, chosen from the twenty-six letters of the alphabet?

A) 14950 **B)** 23751
C) 358800 **D)** 456976

8- How many different committees of 5 people can be chosen from 10 people?

A) 252 **B)** 2002
C) 30240 **D)** 100000

9- Jones is the Chairman of a committee. In how many ways can a committee of 5 be chosen from 10 people given that Jones must be one of them?

A) 126 **B)** 252
C) 495 **D)** 3024

10- A special type of password consists of four different letters of the alphabet, where each letter is used only once. How many different possible passwords are there?

A) 426 **B)** 456976
C) 14950 **D)** 358800

11- A password consists of two letters of the alphabet followed by three digits chosen from 0 to

9- Repeats are allowed. How many different possible passwords are there?

A) 492804 **B)** 650000
C) 676000 **D)** 1757600

Q2

Three-letter words are generated randomly from the five characters a, b, c, d, e, where letters can be used at most once. How many three-different-letter words can be formed from these letters?

Q3

In how many ways can we form a 3-digit number from $\{1, 4, 6, 8\}$ that is less than 600 and greater than 200?

Q4

How many different letter arrangements can be made from the letters in the word “Probability”?

Q5

How many ways are there to choose a committee of 4 persons from a group of 10 persons, if one is to be the chairperson?